

AI MENTAL HEALTH

ANA – 2

Report on “Demographic breakdown analysis”

Dataset Used: Adult Psychiatric Morbidity Survey (APMS) 2014

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1. Introduction:

The Adult Psychiatric Morbidity Survey (APMS) 2014 is a primary data source for monitoring trends in the mental health.

The aim of this analysis is to compare the prevalence of mental health severity levels across different demographic groups, mainly focusing on age, gender, ethnicity.

A Common Mental Disorder is defined by symptoms that cause significant distress and interfere with daily functioning, typically identified through the Clinical Interview Schedule-Revised (CIS-R). The primary objective of this demographic breakdown is to analyze how these disorders vary across the following key dimensions:

- **Age Band:** Identifying life stages where individuals are most vulnerable to mental health symptoms.
- **Gender:** Examining the established disparities in prevalence between men and women.
- **Ethnicity:** Highlighting variations in mental health outcomes across different ethnic backgrounds.

2. Dataset Description:

The dataset used in this analysis is taken from the Adult Psychiatric Morbidity Survey (APMS) 2014, conducted in England. It provides detailed statistical tables focusing on the prevalence and severity of Common Mental Disorders (CMDs) across various demographic groups.

Instead of specific disorder names, the dataset uses CIS-R score categories to represent severity levels:

- 0–5 → No or minimal disorder
- 6–11 → Mild disorder
- 12–17 → Moderate disorder
- 18 or more → Severe disorder
- 12 or more → Clinically significant disorder

For this analysis, the groups were considered:

- Gender
- Age band
- Ethnicity

3. Demographic Analysis:

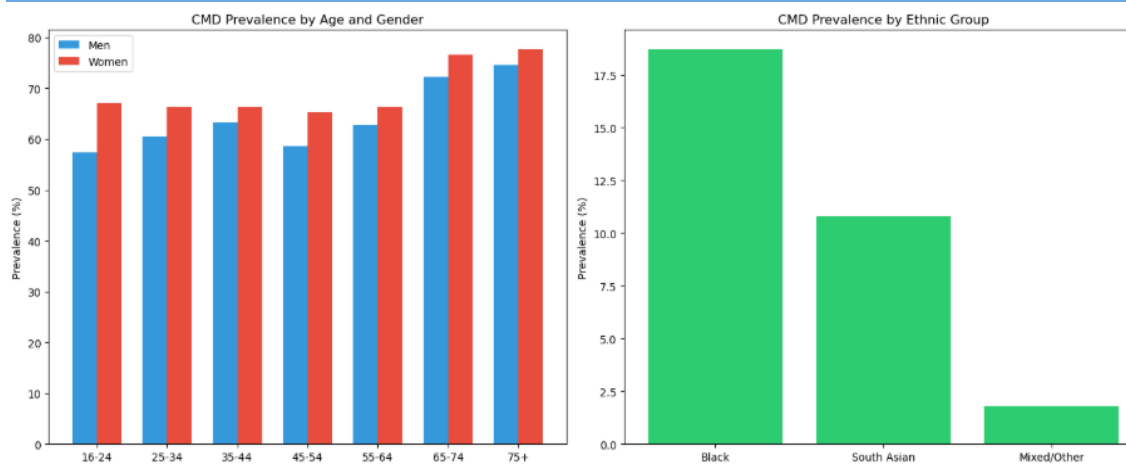
Demographic charts are visual tools used to represent the characteristics of a specific population, such as age, gender, ethnicity, income, or education level.

```
# 4. Visualization
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))
# Plot A: Grouped Bar Chart for Age and Gender
x = np.arange(len(age_groups))
width = 0.35
ax1.bar(x - width/2, men_cmd, width, label='Men', color='#3498db')
ax1.bar(x + width/2, women_cmd, width, label='Women', color='#e74c3c')

ax1.set_title('CMD Prevalence by Age and Gender')
ax1.set_xticks(x)
ax1.set_xticklabels(age_groups)
ax1.set_ylabel('Prevalence (%)')
ax1.legend()

# Plot B: Bar Chart for Ethnicity
ax2.bar(ethnic_groups, ethnic_values, color='#2ecc71')
ax2.set_title('CMD Prevalence by Ethnic Group')
ax2.set_ylabel('Prevalence (%)')

plt.tight_layout()
plt.show()
```

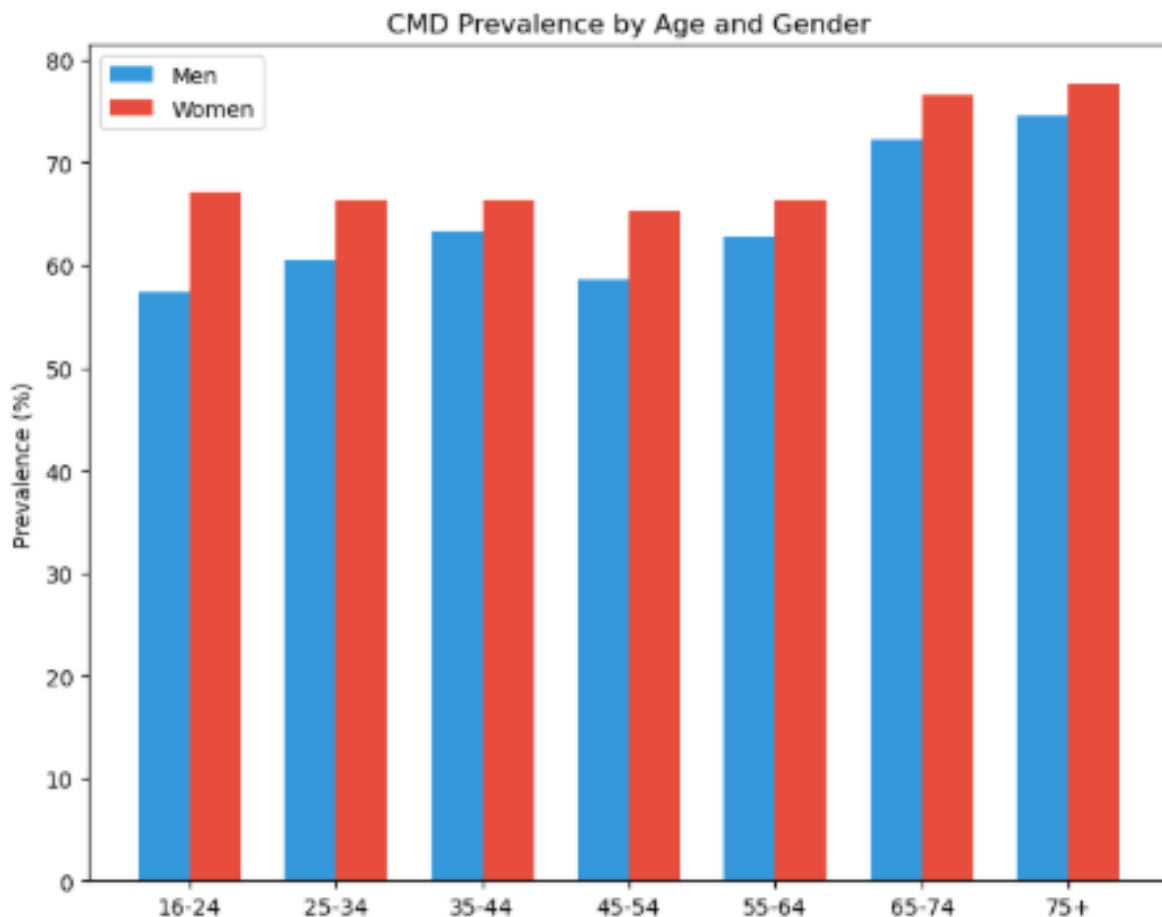


1. Analysis by Age Band and Gender :

The data reveals a consistent trend where women report higher prevalence rates of CMD symptoms across all age categories.

- **Gender Comparison:** In the youngest bracket (16-24), women (67.1%) significantly outpace men (57.3%) in symptom reporting.
- **Peak Prevalence:** The highest rates for both genders are found in the oldest age bands (65-74 and 75+), with women reaching a peak of 77.6% and men 74.6%.
- **Stabilization:** Prevalence rates for women remain relatively stable through middle age (approx. 65-66%), whereas men show more fluctuation before increasing in later life.

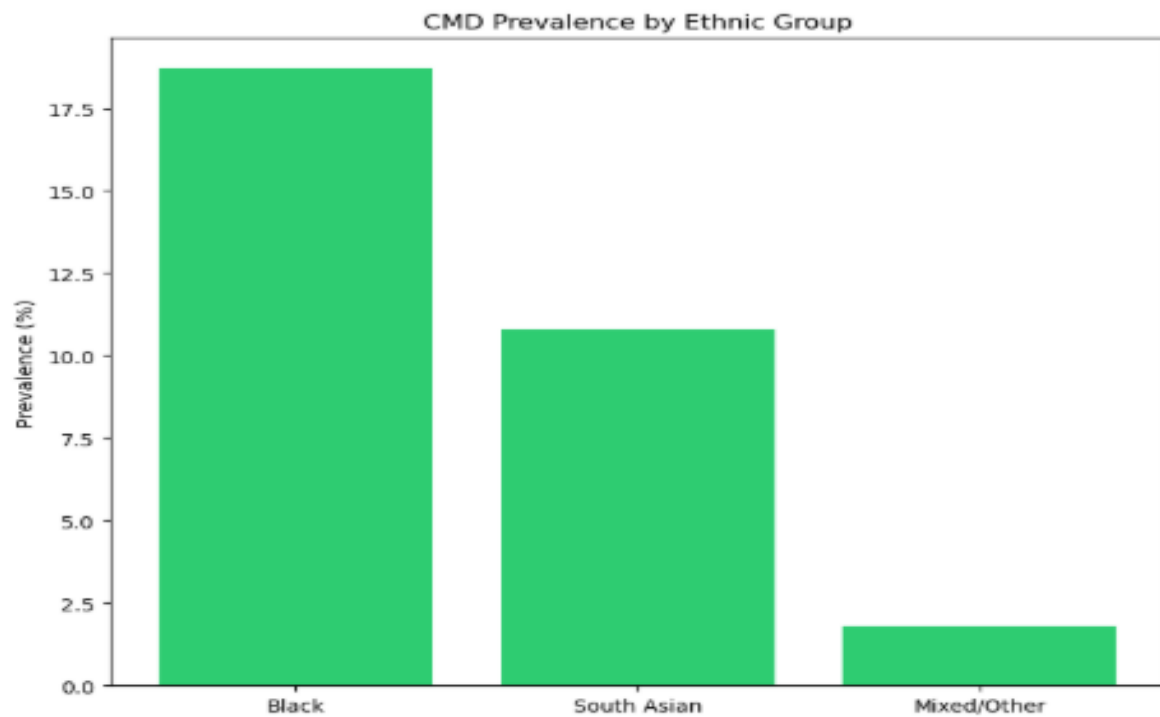
A grouped bar chart was created to compare mental health severity across Age band and Gender.



2. Analysis by Ethnicity

CMD prevalence shows notable disparities when broken down by ethnic background (based on CIS-R scores of 12 or more):

- **Highest Prevalence:** The Black ethnic group reported the highest prevalence at 18.7%.
- **Moderate Prevalence:** The South Asian group followed with a prevalence of 10.8%.
- **Lowest Prevalence:** The Mixed/Other category reported significantly lower clinical symptoms at 1.8%



4. Conclusion

The demographic analysis Adult Psychiatric Morbidity Survey (APMS) 2014 provides critical insights into the distribution of Common Mental Disorders (CMDs) within the adult population. By examining age, gender, and ethnicity, several clear patterns emerge that are essential for tailoring mental health support:

- **Gender and Age Vulnerability:** The data confirms that women consistently experience higher rates of CMD symptoms across all life stages. Of particular note

is the high prevalence in young women (16–24) and the significant increase in symptoms for both genders in the 75+ age bracket.

- **Ethnic Disparities:** Significant variations exist between ethnic groups, with Black adults reporting the highest prevalence of clinical CMD symptoms (18.7%). This highlights the need for culturally sensitive mental health interventions and further investigation into the social determinants affecting these communities.
- **Systemic Prevalence:** With a broad majority of the population reporting some level of CMD symptoms (CIS-R score 6+), mental health concerns appear to be a normative rather than an exceptional part of the adult experience in England.